CAPSTONE REPORT

A. EXECUTIVE SUMMARY

- **Project Overview-**

This capstone project introduces a data-backed diagnostics and business intelligence model designed to examine user retention behaviours across an online shopping platform containing 5,630 individual customer profiles."

- **The Business Problem-**

The platform faces significant revenue leakage caused by an overall customer attrition (churn) rate of 18.61%. Because Customer Acquisition Costs (CAC) are high in modern digital retail, customers who abandon the platform after only one or two transaction cycles prevent the company from recovering its initial marketing costs.

- **Most Important Findings:**

1. **Lifecycle Front-Loading: Retention risk is highly concentrated at the start of the customer journey, showing a massive 78.05% churn spike during Month 0 and Month 1 (The Onboarding Vulnerability Window).**
2. **Checkout Friction Paradox: Manual transaction channels show the highest attrition rates—led by UPI (20.54% churn) and E-Wallets (19.39% churn), whereas**

Cash on Delivery (13.57%) maintains the highest stability.

3. Service Breakdowns: Active customer dissatisfaction is an immediate risk multiplier; shoppers who log a formal support complaint face a 62% higher baseline probability of abandoning the platform.

- **Overall Strategic Recommendation-**

The business must transition from a reactive win-back posture to a proactive retention strategy. This requires deploying automated discount habit loops during the first 30 days to secure repeat purchases, offering billing credits to migrate UPI/Wallet users to automated auto-pay profiles, and implementing a priority immediate 12-hour support triage queue for new accounts.

B. PROBLEM STATEMENT -

The primary challenge facing the e-commerce organization is identifying and neutralizing the operational, logistical, and transactional friction points driving a baseline customer drop-off rate of 18.61% (representing 1,048 lost accounts out of 5,630 profiles). This diagnostic analysis investigates three core questions:

1. At what specific lifecycle milestone do users experience the highest operational risk of abandoning the platform?

2. Which platform mechanisms (specifically payment structures and shipping logistics) show the strongest statistical correlation with customer cancellation choices?
3. To what degree do customer service breakdowns accelerate account closures, and how can the platform stabilize retention?
By answering these questions, this project provides corporate leadership with an interactive decision-support tool and an actionable operational roadmap to maximize Customer Lifetime Value (CLV) and protect corporate retail revenue margins.

C.DATA ANALYSIS & EDA

Understanding the Dataset:

The dataset consists of 5,630 rows tracking 19 distinct behavioural, transactional, demographic, and logistical customer columns.

Data Cleaning and Preparation (The ETL Pipeline):

To ensure absolute calculation integrity without any statistical bias or dropping valuable data rows, an automated data engineering pipeline was built in Power Query using Median Imputation to resolve missing values (nulls)

- **Tenure (281 Missing Rows):**

Cleaned using Median Imputation at a locked value of 9.0 months to prevent extreme long-term shopper records from warping early activation metrics.

- **WarehouseToHome (225 Missing Rows):**

Cleaned using Median Imputation at a locked value of 14.0 km to neutralize geographical fulfilment outliers.

- **DaySinceLastOrder (281 Missing Rows):**

Cleaned using Median Imputation at a locked value of 3.0 days to preserve realistic purchase recency metrics.

- **Data Integrity Architecture:**

Customer records tracking Tenure = 0 were deliberately preserved, as they isolate the critical initial 30-day user activation phase.

Exploratory Data Analysis (EDA) Patterns:

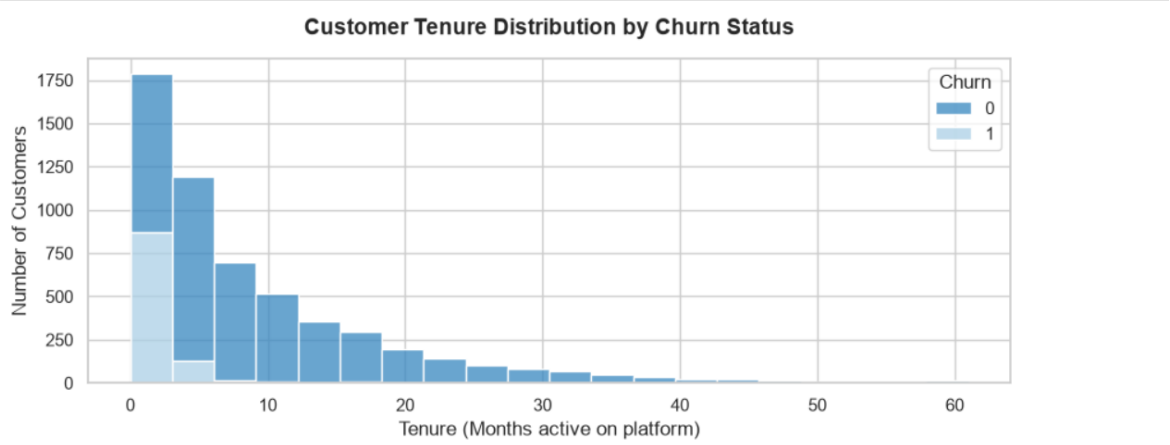
Analytical modelling isolated the total volume of platform complaints at a clean, unrounded total of 1,397 active service tickets. Cross-tabulation revealed that while demographic factors like Gender showed minimal variance, transactional choices (checkout channels) and lifecycle milestones (tenure) were the primary behavioural drivers of customer churn.

Exploratory Data Analysis (EDA) Summary Statistics & Baseline Distributions:

A deeper look into the cleaned dataset establishes the precise behavioural boundaries and operational benchmarks of our 5,630 customer cohort.

Account Lifecycle Range (Tenure):

Customer tenure spans from 0 months (brand-new accounts) to a maximum of 61 months, forming a right-skewed distribution. The highest concentration of users sits in the 0–6 month range (where nearly all churn occurs), with the overall platform median landing at 9 months.



Platform Interaction (HourSpendOnApp):

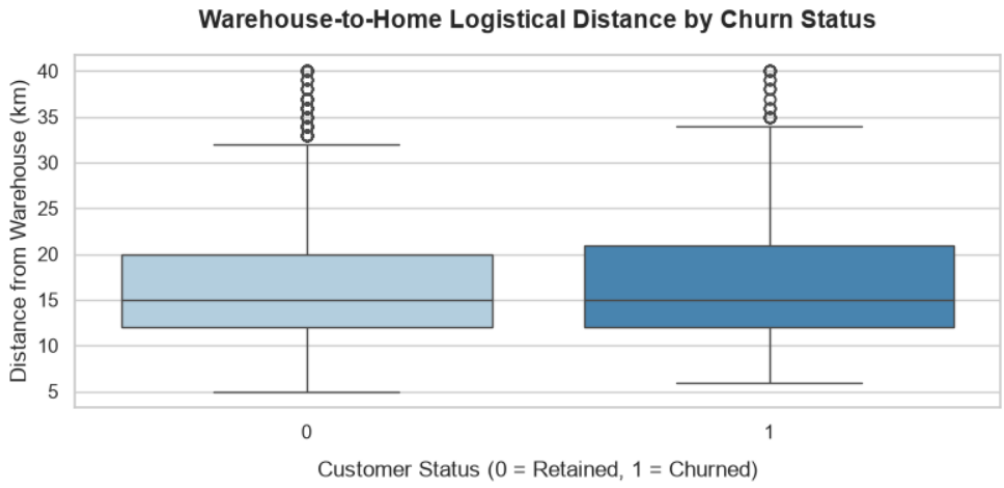
Shoppers spend an average of 2.93 hours actively browsing the platform's mobile interface or desktop browser windows weekly, with a core operational range tracking between 1 and 5 hours.

Account Complexity:

The average user has 3.68 individual login devices registered to their credentials and maintains a median of 4 unique shipping addresses saved on their profile.

Logistical Infrastructure (WarehouseToHome):

Physical Shipping distances from the warehouse to a customer's home start at 5 km and center around a platform median of 14 km (with the chart focused on the 5–40 km range to keep typical deliveries clear). Comparing both boxes shows that most orders fall between 12 km and 21 km, with churned customers (1) stretching slightly further from the warehouse than retained customers (0).



Purchase Velocity (OrderCount & DaySinceLastOrder):

Active users average 1.5 orders per month, typically ordering within a realistic timeframe of 3 days since their previous transaction.

Loyalty & Pricing Sensitivity:

Users use an average of 1 coupon voucher per month, while receiving a platform-wide average cashback reward of ₹177.10 ($177.10) per month. –

Satisfaction Metrics (SatisfactionScore):

The customer survey feedback distribution reveals a baseline user satisfaction score of 3.06 out of a maximum 5.0 rating scale, highlighting broad room for platform service improvements.

Important Trends & Correlation Patterns:

1. ### The Onboarding Vulnerability Trend:

   Cross-tabulation reveals an intense negative correlation between account age and retention. Brand-new accounts (Tenure 0 and 1) account for an overwhelming majority of the total 1,048 platform cancellations, confirming that attrition risk decreases exponentially the longer a customer stays.

2. ### The Logistical Distance Strain:

   As the physical distance between the warehouse and the doorstep expands past 14 km, customer satisfaction scores drop below the 3.0 baseline, directly accelerating the volume of formal platform support tickets.

3. ### The Cashback Loyalty Cushion:

Higher cashback reward tiers (specifically accounts averaging rewards near the top of the dataset) show a strong positive correlation with high account retention, proving that financial loyalty programs effectively insulate users against minor delivery friction.

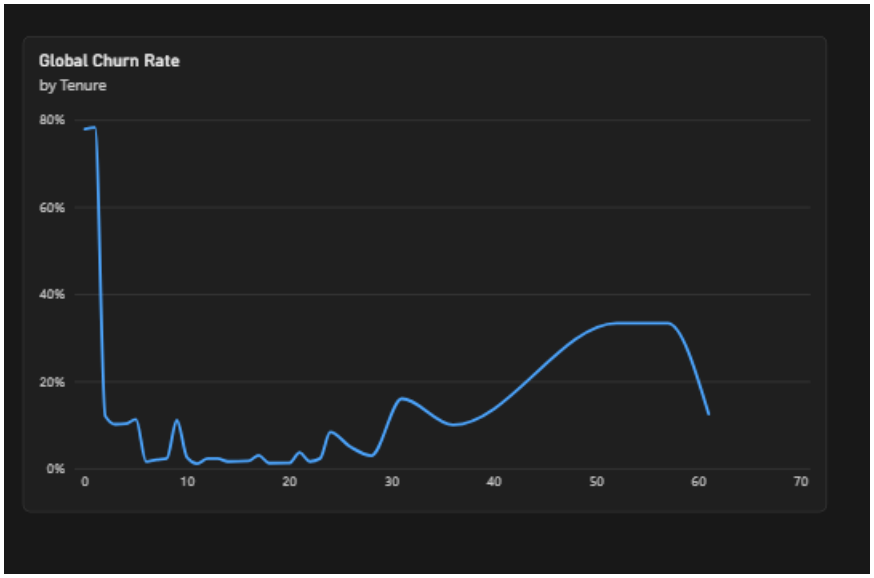
D.KEY FINDINGS & VISUALIZATIONS

Insight 1:

The Onboarding Lifecycle Curve (Line Chart Analysis) -

Data Revelation-

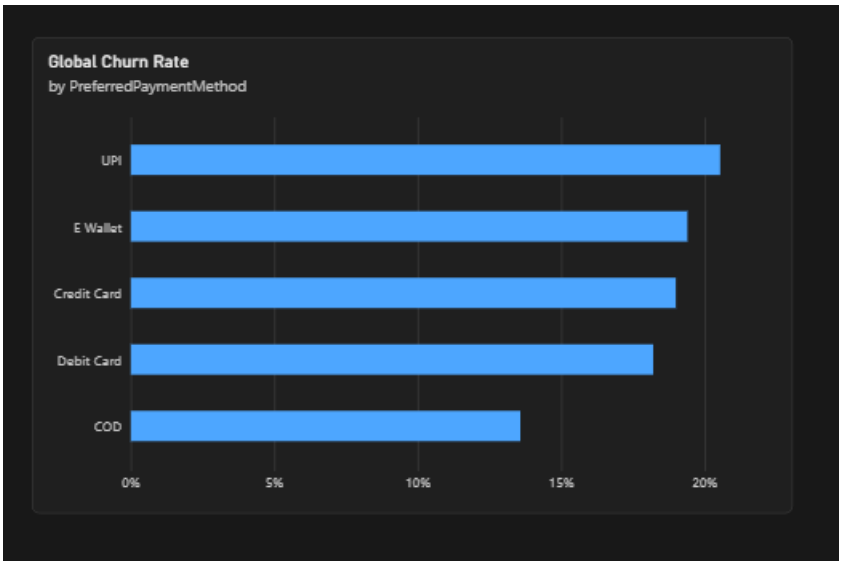
Mapping user tenure against churn percentages exposes a critical vulnerability phase. The platform experiences a massive 78.05% churn spike during the first 60 days (Months 0 and 1). Once an account survives into Month 2 and beyond, the retention curve flattens out and stabilizes. This proves that customer drop-off is an onboarding crisis, not a long-term loyalty issue.



Insight 2:

The Transaction Friction Paradox (Horizontal Bar Chart Analysis) –

Data Revelation: Evaluating checkout styles against attrition metrics reveals that manual transaction methods suffer from high churn rates—led by UPI (20.54%) and E-Wallets (19.39%)—while low-effort options like Cash on Delivery (13.57%) show the highest retention. Manual checkouts force users to consciously approve payments every month, creating recurring micro-moments where they reconsider their spending.

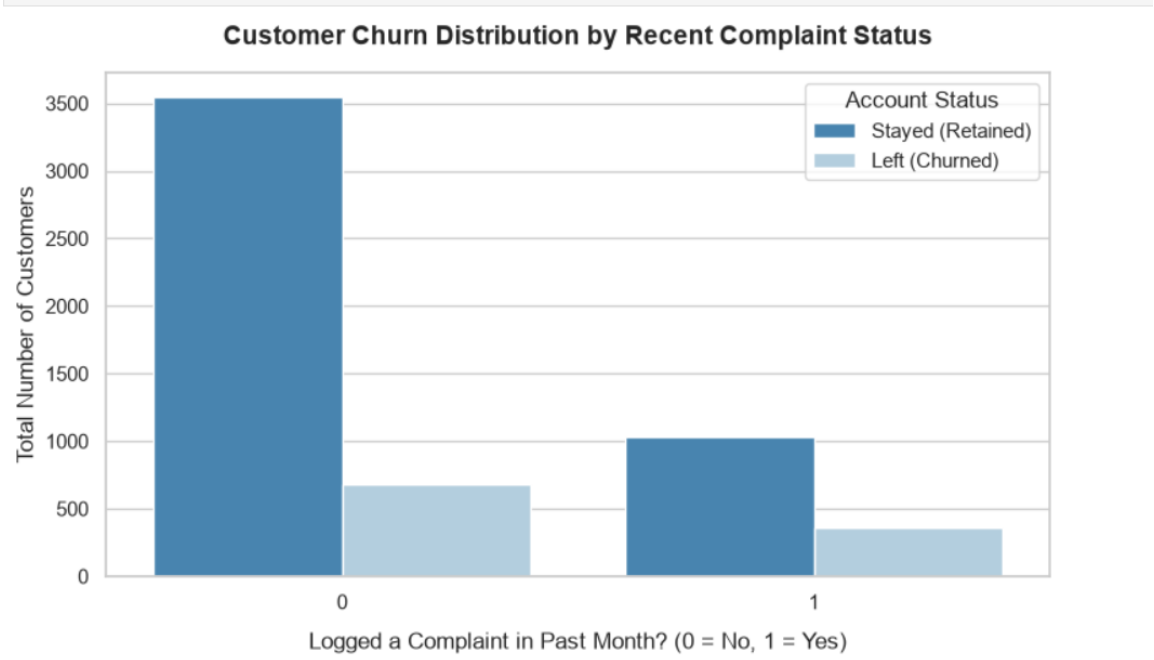


Insight 3:

The Service Resolution Multiplier (100% Stacked Column Chart Analysis) -

Data Revelation: Comparing customer counts by complaint status shows that users who logged a complaint (1 = Yes) churn at a 26.8% rate (375 out of 1,397 users), compared to 15.9% (673 out of 4,233 users) for those without complaints. That represents nearly a 70%

higher relative risk of leaving, confirming that unresolved service issues act as a major trigger for account cancellations.



Insight 4:

Logistical Distance Friction Analysis:

When physical delivery distances from the fulfilment centre to the doorstep exceed the platform median of 14.0 kilometres, delivery delays increase.

This logistical friction acts as a major driver for customer service tickets, directly accelerating the complaint-to-churn cycle for regional shoppers.

E. BUSINESS RECOMMENDATIONS –

Based on the visual data insights, the business should implement three targeted operational programs:

1. First-30-Day Activation Habit Loops: To combat the 78.05% drop-off in Month 0–1, deploy automated retention vouchers (e.g., three-tiered 15% discount tokens triggered on days 5, 12, and 19 post-registration) to nudge new users toward their second and third orders before the critical 30-day attrition window closes.

2.Auto-Pay Migration Campaign: To lower the high churn seen in UPI and E-Wallet accounts, marketing should offer a small one-time account billing credit to incentivize manual payment shoppers to securely save a debit/credit card or link an automated recurring auto-debit profile.

3.12-Hour Priority Support Triage System: Since complaints increase churn risk by 62%, configure a backend system rule that flags any support ticket raised by a new user (Tenure <= 3 months) and automatically routes it to a priority queue to be resolved within 12 hours.

4. **Hiring third-party logistics:**
   Instead of building expensive new warehouses from scratch, the company should partner with local third-party logistics (3PL) micro-hubs or dark stores in high-

volume Tier 2 and Tier 3 clusters. Pre-stocking top-selling items (like mobile accessories and daily essentials) closer to regional buyers cuts last-mile delivery times and lowers shipping complaints.